

Optimization and Curve Sketching



Critical points and the first/second derivative tests

- 1 Find the critical points of $f(x) = x^3 - 6x^2 + 9x$, and classify each as a local max or min using the second derivative test.
 - 2 For $f(x) = x^4 - 4x^3$, find the intervals of concavity and any inflection points.
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Optimization applications

- 3 A farmer has 200 m of fencing to enclose a rectangular field against an existing wall (so only 3 sides need fencing). Find the dimensions that maximize the enclosed area.
 - 4 An open-top box is made from a 12 cm square sheet by cutting equal squares of side x from each corner and folding up the sides. Find the value of x that maximizes the box's volume.
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More optimization

- 5 A cylindrical can must have volume 500π cm³. Using $V = \pi r^2 h$ and surface area $S = 2\pi r^2 + 2\pi r h$, find the radius that minimizes the material used (surface area).
 - 6 Find two positive numbers whose sum is 20 and whose product is as large as possible.
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Curve sketching

- 7 For $f(x) = \frac{1}{3}x^3 - x^2 - 3x + 2$, find the intervals of increase and decrease.
 - 8 Using the results above, classify $x = -1$ and $x = 3$ as local maxima or minima via the first derivative test.
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Minimizing distance and cost

- 9 Find the point on the parabola $y = x^2$ closest to the point $(0, 3)$, by minimizing $D^2 = x^2 + (x^2 - 3)^2$.
 - 10 A rectangular box with a square base and open top must have volume 32 m³. Find the base side length that minimizes the surface area, using $S = x^2 + \frac{128}{x}$ (derived from $V = x^2 h = 32$).
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Sketching using derivative information

- 11 For $f(x) = 3x^4 - 4x^3$, find all critical points and classify each using the second derivative test.
- 12 For $f(x) = xe^{-x}$, find the critical point and determine whether it is a maximum or minimum.